



MANIPAL ACADEMY OF HIGHER EDUCATION

Manipal Institute of Technology, Bengaluru

BTech VI Semester

End Semester Examination - April / May -2025

MOS DEVICE MODELLING [ECE-4406]

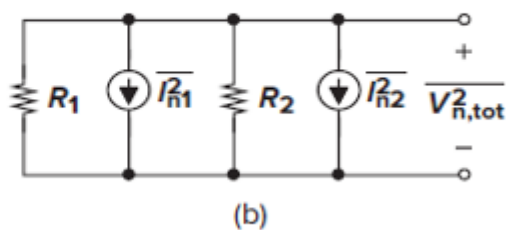
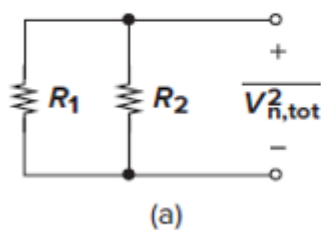
Marks: 50

Duration: 180 mins.

PART-A

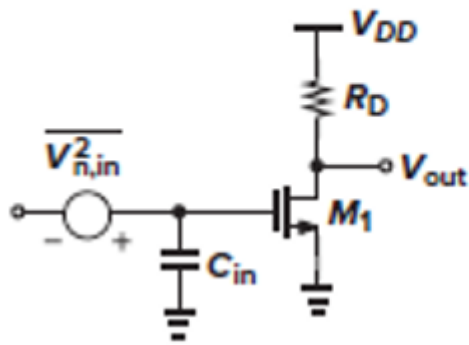
Answer all the questions.

- 1) Explain how the inclusion of velocity saturation and channel length modulation in the Level 2 model affects the simulation results compared to the Level 1 model. (4)
 - A)
 - B) Interpret all the SPICE models with an example with SPICE code (for Diode and MOSFET) (3)
 - C) Implement MOSFET-based circuit to perform DC analysis using SPICE and interpret how these parameters influence circuit performance. (3)
- 2) Analyze the impact of different process corners on the performance of a CMOS circuit. How would you ensure robust design across these variations (5)
 - A)
 - B) Calculate the equivalent noise voltage and current of two parallel resistors R_1 and R_2



- 3) (5)
 - A)

Calculate the input referred noise voltage and current for the circuit shown in figure, including only the thermal noise of M1 & RD.



- B) Model a CMOS inverter using BSIM3 MOSFETs in SPICE (5)
- 4) Examine the key factors influencing strong inversion potential along with its effect on parameters. (5)
- A)
- B) Determine the drain terminal current in BSIM4 with various components in different regions of operation in BSIM3 model. (5)
- 5) Explain why the bias dependence of source/drain resistance is crucial in short-channel MOSFETs used in high-frequency circuit designs. How would ignoring this effect impact the accuracy of circuit simulation results? (4)
- A)
- B) Explain how layout-dependent parasitics contribute to performance degradation. How would you modify the layout or simulation model to minimize these parasitic effects and improve the performance of the circuit? (3)
- C) Explain how model-supported circuit simulations can be used to analyze effects like parasitic elements, power dissipation, and noise. (3)

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